

Facility Operations Platform: Future Advancements and Integration Research

Prepared for: NDWH & CBS Facility Operations Platform

Date: 2026-04-20

1) Executive Summary

Your platform should evolve from a "ticket and inventory tool" into a **Health Infrastructure Intelligence Layer** that predicts failures, recommends interventions, and quantifies county-level service continuity impact.

The most strategic path is:

1. Build a **Facility Risk Engine** (predictive, explainable health score).
2. Add **Optimization and Automation** (routing, triage, next-best-action).
3. Add **Interoperability and Trust** (FHIR-aligned exchange, observability, auditability).
4. Productize as a **County Command Center** (decision support, budget simulation, performance contracting).

2) Product Vision Beyond "Just Inventory"

North-star product identity

- Health Infrastructure Nerve Center
- Facility Digital Twin + Risk Forecasting
- Intervention Intelligence and Cost-Avoidance Analytics

Core strategic outcomes

- Reduce unplanned downtime of critical digital infrastructure.
- Improve mean-time-to-resolve (MTTR) and first-time-fix rate.
- Prioritize interventions by expected risk reduction per budget.
- Provide auditable performance data for county leadership and partners.

3) Future Feature Blueprint

A. Predictive Reliability Layer

- Facility Health Score (0-100) using ticket recurrence, asset profile, connectivity indicators, and historical stability.
- Subcounty Risk Heatmap with 7/30/90-day trend lines.
- Failure early-warning alerts with confidence level and top contributing factors.

- Recurrence intelligence ("similar failure pattern seen in X facilities").

B. Intervention Intelligence

- AI-generated next-best-action recommendations.
- Intervention effectiveness tracker (before/after risk and downtime).
- Playbooks by issue archetype (network, power, endpoint, hybrid).
- Escalation decision support based on impact radius.

C. Optimization and Field Operations

- Technician route optimization for daily assignments.
- Work batching by geography and issue similarity.
- Priority queue sorted by patient-service impact, not only ticket age.
- Offline-first field capture (photos, signatures, geo-time stamps).

D. Executive and Investor-facing Capabilities

- County command brief auto-generated weekly.
- SLA and reliability scorecards by subcounty/facility category.
- Cost-of-downtime and cost-avoidance dashboard.
- Budget simulation ("if we fix top 20 high-risk facilities").

4) Recommended Libraries and Tech Integrations

4.1 Geospatial Intelligence Stack

deck.gl (high-scale, GPU visualization)

- Best use: large-scale subcounty/facility overlays, risk layers, route visualization.
- Why: performant rendering for large datasets, extensible layered architecture.
- Website: <https://deck.gl/docs>

Turf.js (geospatial analytics functions)

- Best use: proximity analysis, catchment polygons, distance buffers, hotspot clustering pre-processing.
- Why: modular GeoJSON-native toolkit, good for in-browser analytics.
- Website: <https://turfjs.org/>

MapLibre GL JS (open-source mapping runtime)

- Best use: map rendering with long-term open-source flexibility.
- Why: community-governed, extensible, WebGL performance, Mapbox-style compatibility.

- Website: <https://maplibre.org/projects/gl-js/>

4.2 Forecasting, Anomaly Detection, and AI Ops

Darts (Python time-series platform)

- Best use: facility-level risk forecasting and anomaly detection pipelines.
- Why: unified API across classical and neural forecasting models.
- Website: <https://github.com/unit8co/darts>

StatsForecast / Nixtla ecosystem (optional add-on)

- Best use: fast baseline forecasting at many-facility scale.
- Why: strong speed and production suitability for large panel time-series.
- Website: <https://nixtlaverse.nixtla.io/>

River (streaming ML; optional)

- Best use: online anomaly detection from streaming telemetry/events.
- Why: incremental learning for real-time adaptation.
- Website: <https://riverml.xyz/>

4.3 Optimization and Planning

Google OR-Tools

- Best use: technician routing, job scheduling, shift constraints, travel-time optimization.
- Why: mature operations-research engine with VRP support.
- Website: <https://developers.google.com/optimization/routing/vrp>

4.4 Observability, Reliability, and Platform Trust

OpenTelemetry (Node.js)

- Best use: tracing API latency, job pipelines, error hotspots, release regressions.
- Why: standard observability layer for resilient scaling and investor-grade reliability.
- Website: <https://opentelemetry.io/docs/languages/js/getting-started/nodejs/>

Sentry (optional complement)

- Best use: front-end and API error intelligence, release quality monitoring.
- Website: <https://sentry.io/for/javascript/>

4.5 Data and Interoperability

HL7 FHIR-aligned exchange (selective adoption)

- Best use: structured data exchange with partner systems where relevant.
- Why: future-proofs interoperability strategy in health ecosystems.
- Reference landing pages:
- <https://hl7.org/fhir/>
- <https://build.fhir.org/>

PMTiles (optional for distribution)

- Best use: ship large static map tiles efficiently from object storage/CDN.
- Why: reduces tile server complexity and cost for geo-heavy workflows.
- Website: <https://protomaps.com/docs/pmtiles/>

5) Architecture Recommendations (Practical)

5.1 Add a "Risk Engine" service boundary

- Keep current app as operational system of record and UI.
- Add a background analytics service that computes:
- facility_risk_score
- predicted_failure_probability_30d
- intervention_recommendation
- confidence and feature attributions

5.2 Event + feature pipelines

- Build event tables from tickets and asset changes.
- Derive model features daily (or hourly later).
- Cache risk outputs for API/UI low latency.

5.3 Explainability first

- Every risk score must expose "why" (top factors).
- This improves field adoption and leadership trust.

5.4 Governance and data quality

- Add data quality score per facility (missing subcounty, stale assets, inconsistent names).
- Add immutable audit log for critical updates.

6) 12-Month Delivery Roadmap

Phase 1 (0-3 months): Intelligence foundation

- Facility Health Score v1 (rules + simple time-series baseline).
- Risk queue view (top high-risk facilities by county/subcounty).
- Command brief export (weekly PDF/email).
- KPI baseline dashboards (MTTR, recurrence, SLA breach).

Phase 2 (3-6 months): Predictive and optimization

- Forecasting pipeline with anomaly detection.
- OR-Tools route optimizer pilot for field teams.
- Intervention recommendation engine v1.
- Evidence capture on interventions (outcome and impact).

Phase 3 (6-9 months): Automation and scale

- AI triage assistant for ticket categorization and assignment.
- Trigger-based playbook automation.
- Expanded observability (OpenTelemetry + alerting SLOs).

Phase 4 (9-12 months): Strategic productization

- Budget impact simulator.
- County benchmarking and resilience index.
- Partner interoperability packages (FHIR-aligned exports/APIs).

7) High-Value KPIs to Track

- Mean time to resolve (MTTR)
- 30-day recurrence rate
- Predicted vs actual failure hit-rate
- SLA attainment by severity
- Downtime hours avoided
- Cost avoided from preventive intervention
- Facilities with complete data (%)

8) Risks and Mitigations

Risk: model trust is low

- Mitigation: explainable scores, transparent factors, conservative thresholds first.

Risk: data inconsistency across facility names/sources

- Mitigation: canonical facility identity, fuzzy match pipeline, confidence scoring for matches.

Risk: complexity grows too quickly

- Mitigation: phased roll-out, each phase tied to measurable KPI impact.

Risk: platform perceived as generic ticketing

- Mitigation: map-first risk UX, outcome language, command-center narrative.

9) Immediate Next Steps (Actionable)

1. Implement `facility\_risk\_score` data model and nightly job.
2. Add Risk Queue page with filters by location/subcounty/system.
3. Add intervention recommendation card on facility details.
4. Introduce OpenTelemetry tracing in API routes and jobs.
5. Pilot OR-Tools scheduling on one county team.

10) Key Research References

- deck.gl docs: <https://deck.gl/docs>
- Turf.js: <https://turfjs.org/>
- MapLibre GL JS: <https://maplibre.org/projects/gl-js/>
- OR-Tools VRP: <https://developers.google.com/optimization/routing/vrp>
- OpenTelemetry Node.js: <https://opentelemetry.io/docs/languages/js/getting-started/nodejs/>
- Darts time-series library: <https://github.com/unit8co/darts>
- FHIR standards landing pages: <https://hl7.org/fhir/> and <https://build.fhir.org/>